The Burden of Preventable Eye Disease in Rural Peru: Clinical Findings from the HAPIN Eye Substudy

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Abstract

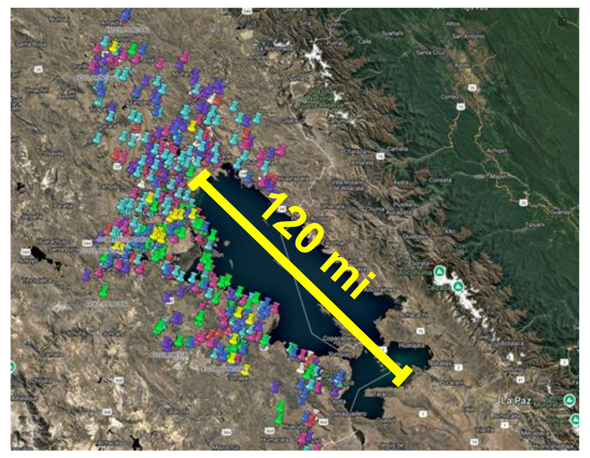
- Limited eye care access among rural communities surrounding Puno, Peru, has contributed to a significant gap in understanding the ocular diseases affecting this population.
- This study aimed to characterize the burden of eye disease in rural Peru.
- Data collected in this study demonstrated significant vision loss due to non-refractive causes[4].
- This finding highlights the potential role of environmental exposures, such as sunlight and household air pollution, in eye disease.
- The ancillary study provides a foundation for future research into the environmental determinants of eye disease in underserved rural communities.

Introduction

- Over 1 billion people globally live with preventable vision loss with rural communities disproportionately affected[1].
- Rural Andean populations face unique risks from: Intense UV exposure, household air pollution due to traditional cooking practices and limited access to eye care.
- The elevation in Puno, Peru, reaches approximately 13,000 feet, resulting in significantly higher UV exposure than at sea level[3].
- The burden of ocular diseases and visual impairment in this population is underreported.
- This study examines the prevalence of eye disease, symptoms, and visual impairment as well as environmental and behavioral risk factors contributing to undiagnosed ocular conditions in rural Peruvian women.

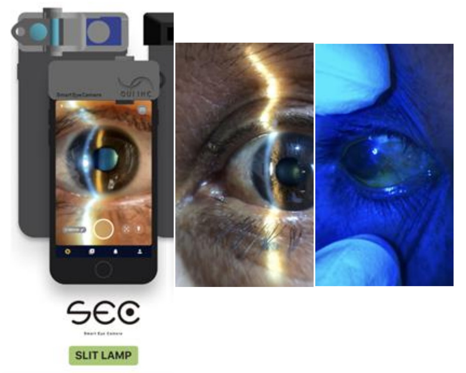
Methods and Materials

Study Design:
Cross-sectional study nested within the Peru site of the *Household Air Pollution Intervention Network (HAPIN)* trial[5].
Study Population:
Rural women aged ≥40 years from Puno, Peru, who were previously enrolled in HAPIN as older adult women.



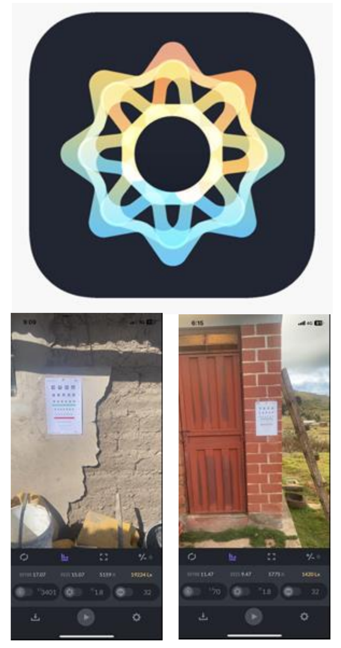
Data Collection:
Eye Health Survey
Collected ocular history/environmental exposure and DEQ5 surveys.

Ocular Examinations
Visual Acuity: Uncorrected and pinhole acuity using Snellen charts or Tumbling E.
Anterior Segment Evaluation: Images of anterior segments captured with portable slit lamp and analyzed by local ophthalmologist Dr. Fredy Santillana Medina.



Environmental Measures:
Light exposure measured in lux using a digital light meter
Behavioral data on hat/sunglasses use and occupational exposures.

Analysis:
Descriptive statistics for symptom prevalence, diagnoses, and visual acuity.
Correlation analyses between environmental factors and visual outcomes.
Comparison of symptom reports with clinical findings to identify undiagnosed disease.



Results

Survey results

Common symptoms:

- Light sensitivity (73%)
- Blurry vision (62%)
- Double vision (61%)
- Floaters (58%)
- Eye pain (40%)

Symptoms often occurred daily or weekly.

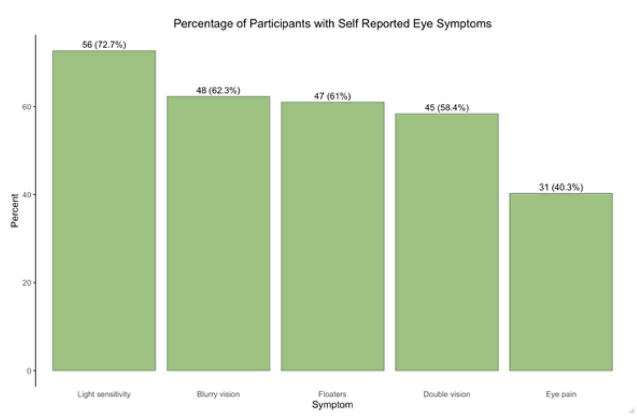
Only **9%** currently use eye drops; **37.7%** used them in the past, indicating significant gap between symptom burden and access to care.

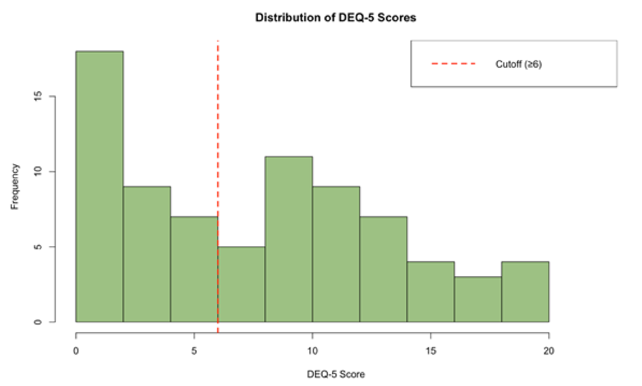
High sunlight exposure: **91% ≥5 hrs/day**.

0% used sunglasses and 88% had no chemical exposure.

Dry eye screening (DEQ-5):

- Mean score: 7.9
- 57% screened positive (≥6)





Eye	Mean Decimal	Approx. Snellen
Uncorrected Right eye (OD)	0.67	20/30
Uncorrected Left eye (OS)	0.64	20/32
Right eye pinhole	0.48	20/42
Left eye pinhole	0.46	20/43

Visual Acuity Results

0% wore corrective lenses
36.4% had vision worse than 20/40 in ≥1 eye

- **16** had bilateral impairment

Pinhole test (n=28):
32% improved to 20/40 in both eyes

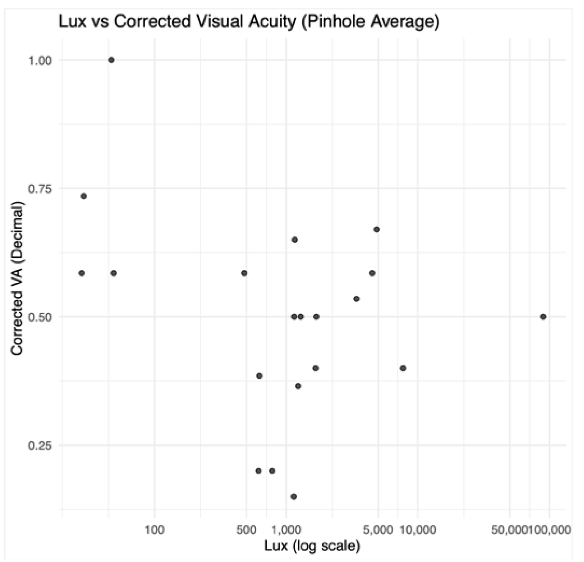
- **9%** improved in one eye

Most impairment likely due to **non-refractive causes** (e.g., cataract, dry eye)

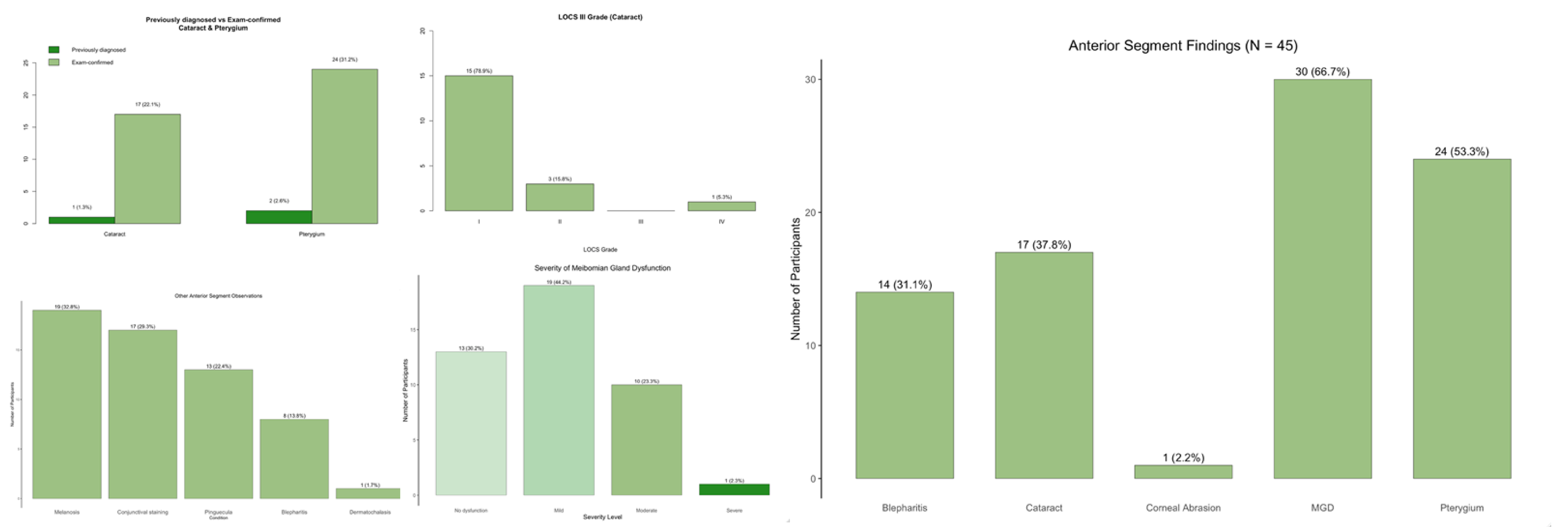
Environmental Exposure Results

Average measured light (lux): 5,978
Range: 28 to 89,511 lux
Recommended range for visual acuity testing: 2,000–10,000 lux
Participants exposed to >10,000 lux: 8
Correlation with corrected visual acuity:

- Spearman's ρ = –0.182
- **p = 0.44**



Anterior Segment Exam Results



Slit lamp exams revealed a high burden of undiagnosed, treatable eye disease, demonstrating the strong diagnostic yield of targeted screening in this rural population.

- Pterygium was present in 53% of participants, with 84% previously undiagnosed.
- Cataracts were found in 38%, 90% of which had not been diagnosed before.
- Meibomian Gland Dysfunction (MGD) 67% had mild MGD, and 11 (18%) had moderate to severe MGD. 21% had no gland dysfunction on exam.
- Other findings included blepharitis (28%) and one symptomatic corneal abrasion.

Importantly, most participants with visual impairment did not improve with pinhole testing, suggesting that vision loss was largely due to non-refractive causes such as cataract and ocular surface disease, rather than uncorrected refractive error.

Discussion

- Slit lamp exams revealed a high prevalence of undiagnosed, treatable anterior segment conditions in this rural population.
- Targeted clinical screening was highly diagnostic for pterygium, cataract, and MGD, highlighting its value in resource-limited settings.
- Pinhole testing confirmed that structural causes of vision loss were common, as most participants did not improve with it. Most vision impairment was due to non-refractive causes.
- There was a mismatch between high symptom burden and low access to treatment (e.g., 0% using sunglasses; only 9% using eye drops).
- Findings support the need for increased access to anterior segment evaluation and eye health services in rural Andean communities.

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